Basic Emergency Care: for Electrical Injury Cases

DR. YAHYA KAMAL MOHAMMEDALI
INTERNIST / CLINICAL ENDOCRINOLOGIST
HAWLER MEDICAL UNIVERSITY – COLLEGE OF MEDICINE

Objectives

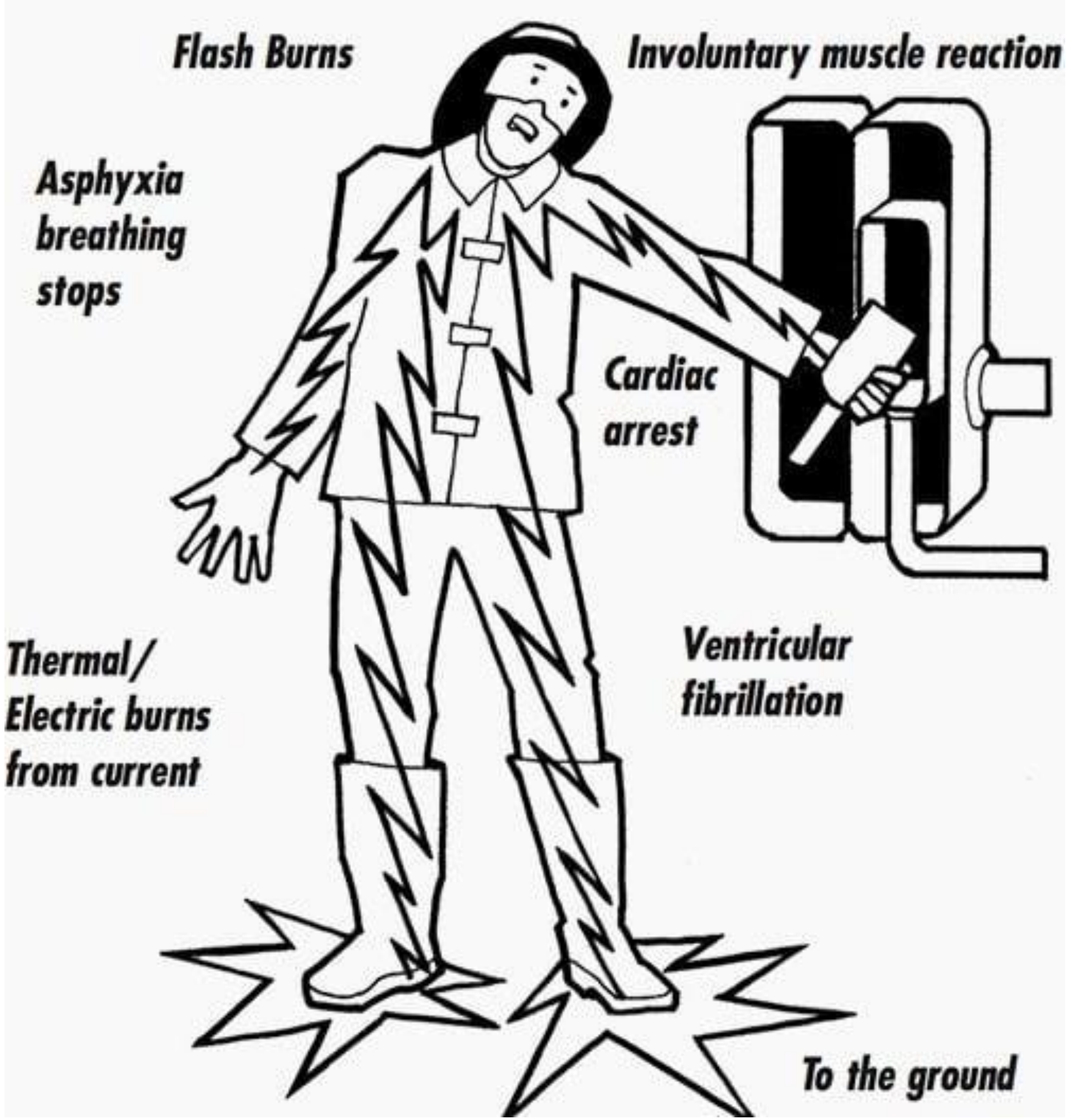
- To familiarize with the dangers of electrical injuries
- To provide the knowledge to respond to electrical injury emergencies
- To emphasize the importance of aftercare and follow-up care for electrical injury patients
- To raise awareness of preventive measures and safety tips

Scope:

- This lecture will provide an understanding of the basic emergency care procedures for electrical injury cases.
- The lecture will cover the dangers of electrical shock, the proper response to electrical burns, and the importance of aftercare and follow-up care.
- The lecture will also discuss preventive measures and safety tips to help reduce the risk of electrical injuries, with a focus on the role of medical professionals in providing adequate care for electrical injury patients.

Electrical injuries

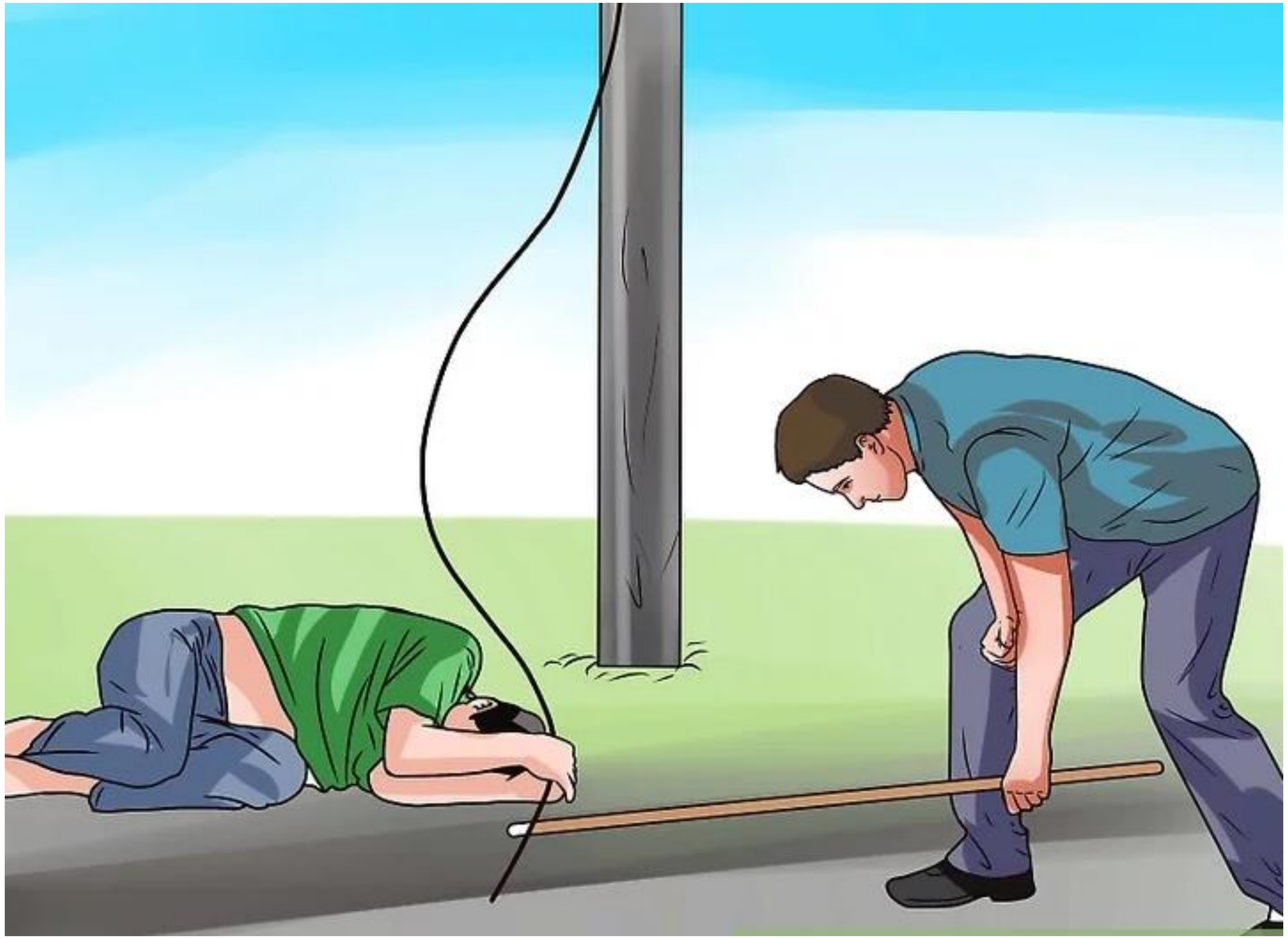
- Types of electrical injuries: Electrical injuries can be classified into two main categories: high voltage and low voltage. High voltage injuries are more severe and can cause deeper burns and more extensive internal damage. Low voltage injuries are less severe but can still be dangerous.
- When a person comes into contact with electrical current, it can cause electrical shock. Symptoms of electrical shock can include muscle contractions, unconsciousness, cardiac arrest and respiratory distress.



Flash Burns
Involuntary muscle reaction
Asphyxia breathing stops
Cardiac arrest
Thermal/ Electric burns from current
Ventricular fibrillation
To the ground

Emergency care for electrical injury cases

1. Safety: Before providing any care, make sure the electrical source has been turned off or that the injured person has been moved away from it.
2. Assessment: Check the person's airway, breathing, and pulse. If they are not breathing, begin CPR. If the person has a pulse but is not breathing, provide rescue breaths.
3. Burns: Electrical burns can be serious, even if they are not visibly apparent. Look for burns at the entry and exit points on the body and any charred or blackened clothing. Cover the burns with a sterile dressing or clean cloth.



Emergency care for electrical injury cases

4. Cardiac arrest: Electrical shock can cause cardiac arrest. If the person has no pulse, begin CPR immediately.
5. Neurological symptoms: Electrical shock can cause seizures, muscle spasms, or unconsciousness. If the person is conscious, keep them still and comfortable. If they are unconscious, place them in the recovery position.
6. Seek medical attention: Electrical injuries can have serious internal effects, so it's important to seek medical attention even if the person appears to be okay.


AMBULANCE
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Emergency care for electrical injury cases

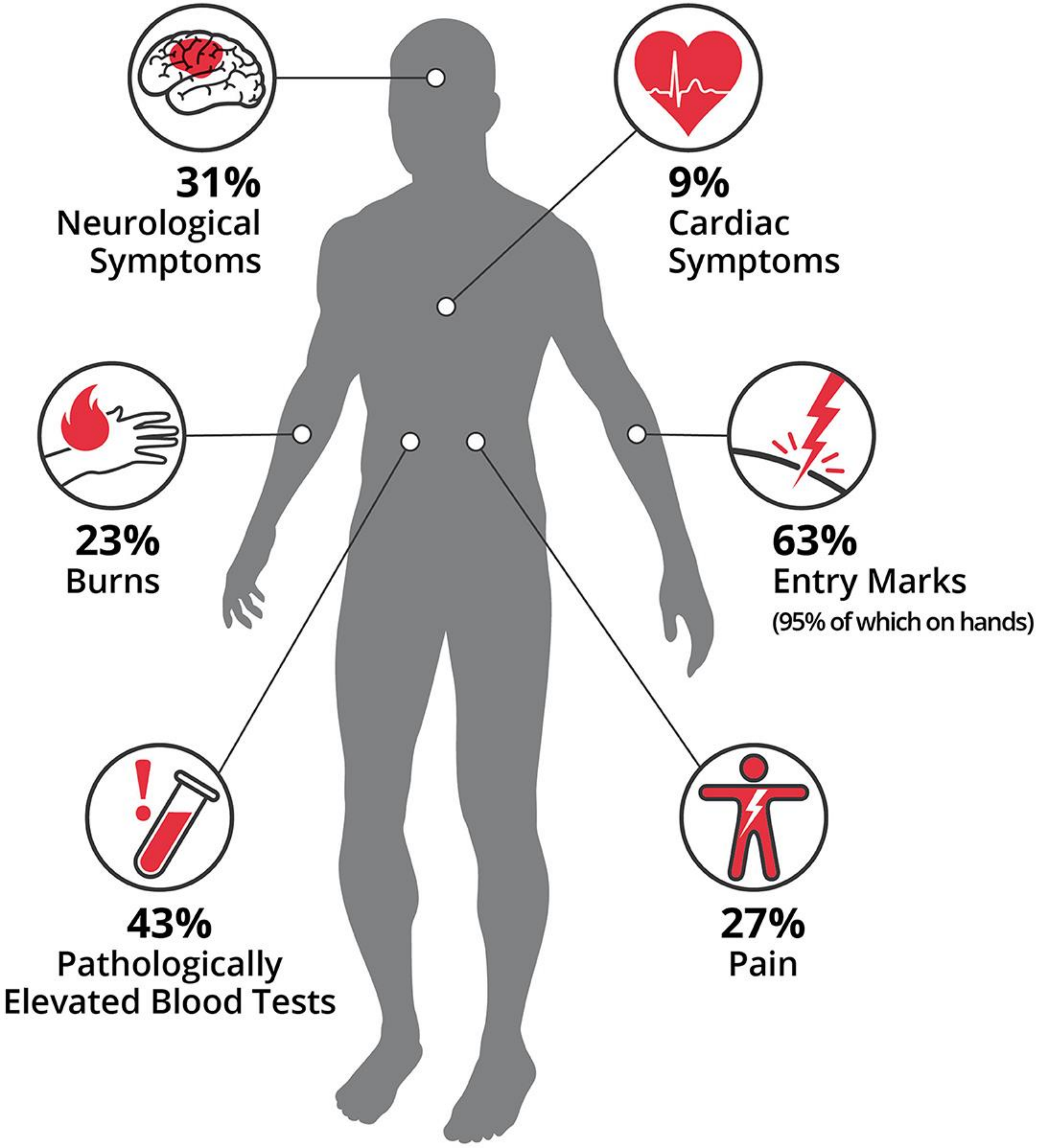
7. Monitor vital signs: Keep an eye on the person's pulse, breathing, and blood pressure. If any of these deteriorate, seek medical attention immediately.
8. Remove jewelry and clothing: Electrical burns can cause swelling, so it's important to remove anything that might constrict the affected area, such as jewelry or tight clothing.
9. Pain management: Burns can be painful, so consider giving the person over-the-counter pain medication if they are in pain

Emergency care for electrical injury cases

10. Electrical burns: Electrical burns can be particularly dangerous because they can cause deep tissue damage that is not immediately visible. Treat electrical burns as serious injuries and seek medical attention as soon as possible.
11. External burns: External burns caused by electrical injury can be accompanied by internal burns, which can be more serious. Seek medical attention even if the external burns appear minor.

Emergency care for electrical injury cases

12. Follow-up care: After initial medical attention has been provided, it's important to follow up with a doctor or specialist to monitor for any delayed effects or complications from the electrical injury.
13. Aftercare: After medical attention has been sought, the person may need further treatment, such as wound care, pain management, or rehabilitation.
14. Also, electrical injury patients should be monitored for potential complications, such as infections and nerve damage


31%
Neurological Symptoms
9%
Cardiac Symptoms
23%
Burns
63%
Entry Marks
(95% of which on hands)
43%
Pathologically Elevated Blood Tests
27%
Pain

CARDIAC

Arrhythmias

- The commonest complication, supraventricular arrhythmias (sinus tachycardia, atrial extrasystoles, or atrial fibrillation) and ventricular arrhythmias (extrasystoles, tachycardia, or fibrillation)[6-8]
- Exposure to high voltage current is more likely to cause cardiac standstill (asystole), but even low voltage alternating current can cause cardiac arrest by ventricular fibrillation[9]
- There is a vulnerable period in every heart beat during which exposure to energy at a frequency of 50-60 Hz, used in most household and commercial electrical sources, can trigger fibrillation
- Although most life threatening events occur immediately after electric shock, delayed ventricular arrhythmias have been reported (up to 12 hours after the incident, with low as well as with high voltages)[10]
- The mechanism behind electrically induced cardiac arrhythmias is hypothesised to be initial damage to heart muscle and subsequent scar formation leading to abnormal electrical activation of the heart

Bradycardia

- Severe slowing of the heart rate can result from interference with the normal electrical (conduction) system, which can be delayed for months or years after the accident[11]
- The sino-atrial and atrio-ventricular nodes responsible for impulse generation and propagation within the heart may be more susceptible to damage by electrical injury than other cardiac cells[12 13] (necropsy shows widespread areas of cell death, mainly in these electrical pathways[14])

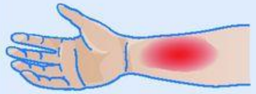
Heart muscle injury

- May result from reduction of blood supply (ischaemia) or direct tissue death (necrosis)
- Chest pain may be absent, and injury may manifest only as non-specific electrocardiographic changes (fig 4), increased levels of myocardial proteins in the blood (troponin) from damaged tissue, or abnormalities such as altered contraction patterns on echocardiography or cardiac MRI[15]
- Occasionally, mostly after high voltage accidents, myocardial infarction can be caused by occlusion of coronary arteries by blood clots or spasm[16]
- Cardiac function usually returns to normal shortly after the event, but cardiac abnormalities sometimes persist

RESPIRATORY ARREST

- Electric current, by interfering with usual transmission of nervous impulses, can paralyse the respiratory muscles (such as diaphragm) or cause them to seize up abruptly (tetanic contraction)
- Inhibition of the centre controlling breathing in the brain

SKIN BURNS

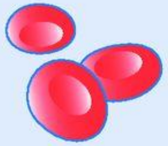


- Extensive burns can result in significant body fluid loss and infection, due to loss of skin barrier

NEUROLOGICAL

- Damage to nerve tissue may cause loss of consciousness, impaired recall, spinal cord injury, paralysis, or loss of sensations in limbs
- Neuropsychological problems are often underappreciated, but post-traumatic stress disorder, depression, and chronic neuropathic pain have been reported[17]

VASCULAR

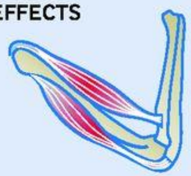


- Clotting in blood vessels due to injury to vessel walls can compromise blood flow to organs

KIDNEY FAILURE

- Myoglobin tubular precipitation caused by damaged muscles releasing myoglobin to the circulation and thus to the kidneys

MUSCULOSKELETAL EFFECTS



- Bone fractures and luxations
- Direct muscle damage, rhabdomyolysis[18]
- Compartment syndrome

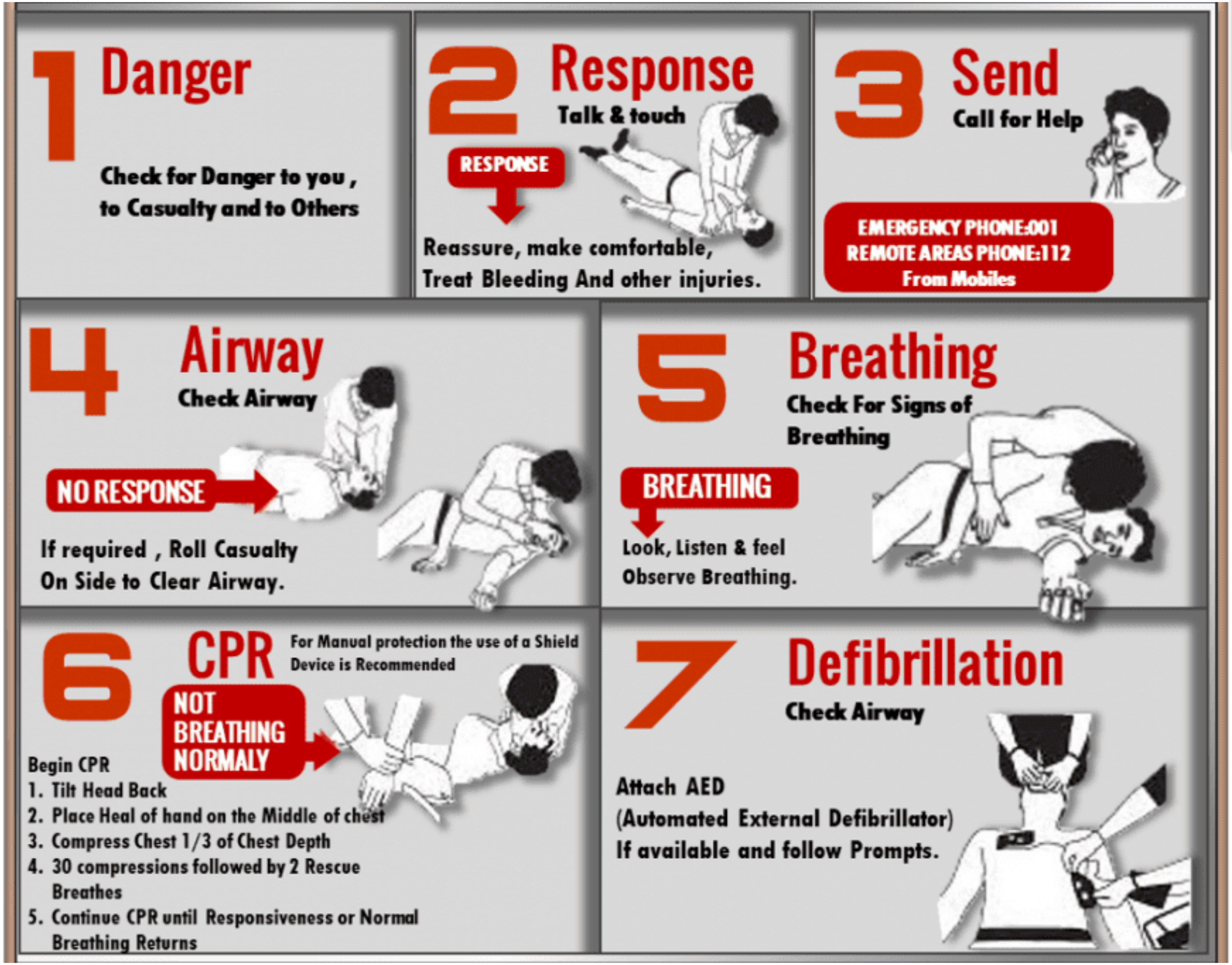
Preventive Measures and Safety Tips

- Follow safety guidelines and use protective equipment when working with electricity
- Make sure electrical equipment is properly maintained and in good working condition
- Consider installing ground-fault circuit interrupters in high-risk areas
- Know CPR and be prepared to respond to an emergency
- Seek training and education on electrical safety from reliable sources


Head protection
Falling or flying objects, overhead objects
Eye protection
Blowing dust or particles, metal shavings, acids or caustic liquids, welding light
Hearing protection
Loud tools and machinery, poorly maintained equipment
High-visibility hat, vest, pants
Errant vehicles, distracted drivers
Chaps pants
Chainsaws
Hand protection
Sharp or hot objects, chemicals, biological or electrical hazards
Foot protection
Falling or rolling objects, sharp or heavy objects, wet and slippery surfaces, uneven surfaces, hot surfaces, electrical hazards
Harness lanyard
Working more than 6 feet or more above a lower level

Key Take Home Messages

- Electrical injury is a serious and potentially life-threatening condition
- Early recognition and proper emergency response is critical for the best outcomes
- Aftercare and follow-up care are important for full recovery and to prevent long-term complications
- Electrical safety and preventive measures should be a priority for both the general public and medical professionals


1 Danger
Check for Danger to you, to Casualty and to Others
2 Response
Talk & touch
RESPONSE
Reassure, make comfortable, Treat Bleeding And other injuries.
3 Send
Call for Help
EMERGENCY PHONE:001
REMOTE AREAS PHONE:112
From Mobiles
4 Airway
Check Airway
NO RESPONSE
If required, Roll Casualty On Side to Clear Airway.
5 Breathing
Check For Signs of Breathing
BREATHING
Look, Listen & feel
Observe Breathing.
6 CPR
For Manual protection the use of a Shield Device is Recommended
NOT BREATHING NORMALY
Begin CPR
1. Tilt Head Back
2. Place Heal of hand on the Middle of chest
3. Compress Chest 1/3 of Chest Depth
4. 30 compressions followed by 2 Rescue Breathes
5. Continue CPR until Responsiveness or Normal Breathing Returns
7 Defibrillation
Check Airway
Attach AED
(Automated External Defibrillator)
If available and follow Prompts.

Key Take Home Messages

- Remember, as a 1st year medical student, you play an important role in providing adequate care for electrical injury patients. Stay informed about the dangers of electrical injuries, the proper response to electrical injury emergencies, and the importance of aftercare and follow-up care. Keep electrical safety and preventive measures in mind, and be prepared to respond in the event of an emergency.

QUESTIONS?